



STUDENT DROPOUT

A Two-Stage Targeting Model for
Caldwell University

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THE PROBLEM & THE EVIDENCE

25.6%

of Caldwell students withdraw — 115 of 450

The decision

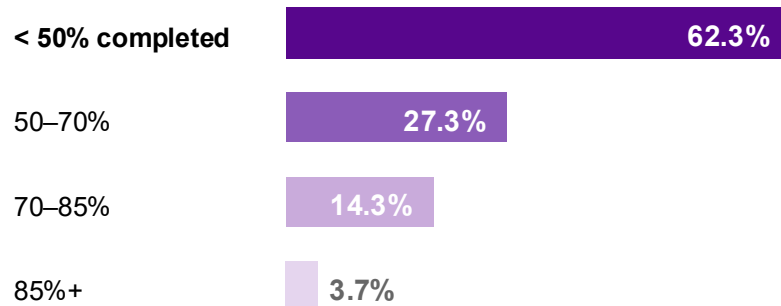
How should the Dean of Student Success allocate finite advising capacity across hundreds of students each term?

*A missed at-risk student costs 30x more than a false flag — **recall over precision.***



Dropout risk by Credit Completion Ratio

Risk concentrates sharply with academic struggle.



A TWO-STAGE APPROACH

STAGE 1

Classification

Random Forest screens which students are at elevated risk. Tuned for **recall** over precision — missing an at-risk student costs 30x more than a false flag.

STAGE 2

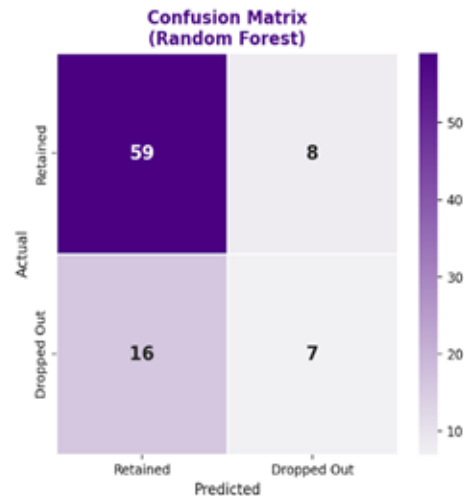
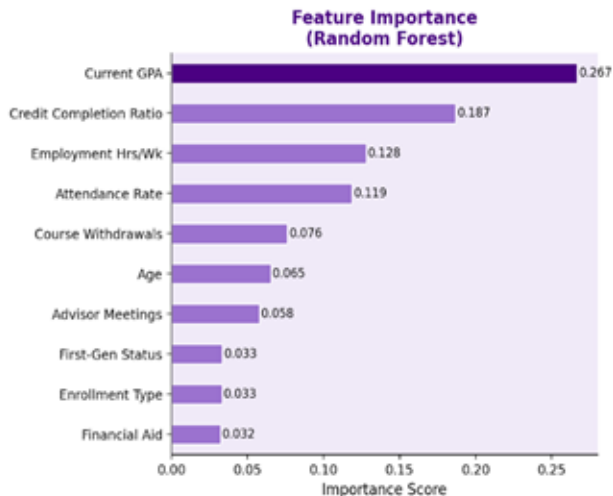
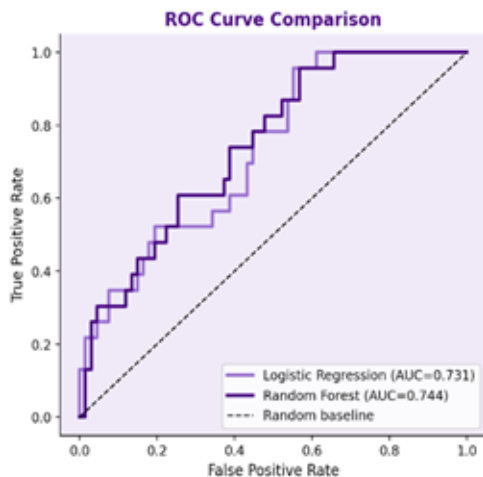
Prediction

Linear regression on the 115 dropouts predicts how far along flagged students will be when they leave. Separates **urgency tiers** so the Dean can match intervention type to expected exit timing.

STAGE 1 RESULTS

Accuracy: 73% | Precision: 47% | **Recall: 30%** ★ optimization target | F1: 0.37 | AUC-ROC: 0.744

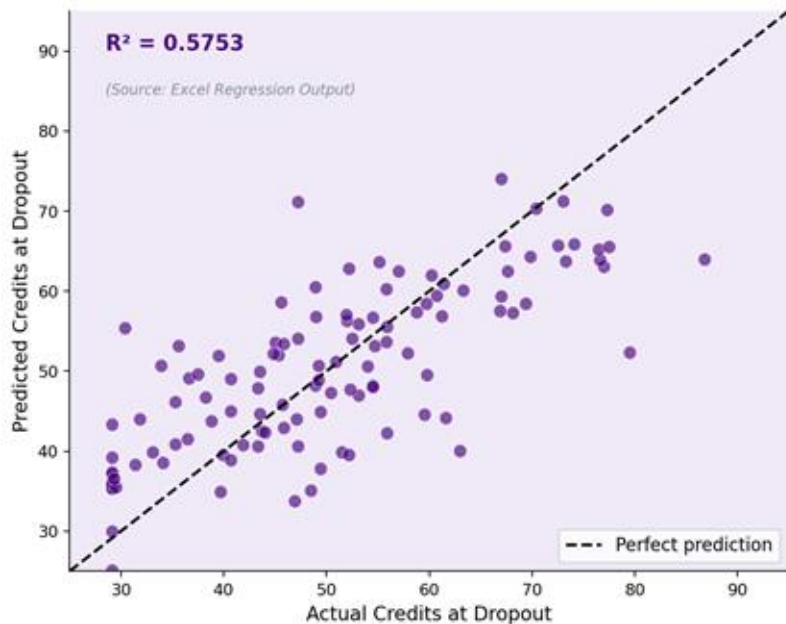
Predictive Modeling Results



AUC = 0.744. Recall = 30% on the dropout class — the optimization target.
Top features: GPA (0.27) and Credit Completion Ratio (0.19) drive 45% of predictive power.

STAGE 2 RESULTS

Actual vs. Predicted Credits at Dropout
(Linear Regression — N=115 Dropout Students)



Two distinct dropout profiles

Splitting at the median (49 credits)

EARLY (n=57)

AVG GPA

2.02

WORK HRS/WK

18.7

→ Academic intervention

LATE (n=58)

AVG GPA

2.47

WORK HRS/WK

25.9

→ Financial / scheduling

Note: Model trained on full dropout sample (N=115), R^2 validated against Excel Analysis ToolPak output.

RECOMMENDATIONS & STAKEHOLDERS

HIGH-TOUCH

WHO

Early-dropout profiles

ACTION

Tutoring, course-load adjustment, probation review

Weekly contact

Early loss > \$15K; fastest ROI

BI-WEEKLY

WHO

Late-dropout profiles

ACTION

Financial aid counseling, flexible enrollment

Bi-weekly meetings

Financial pressure, not GPA

MONITORING

WHO

Lower-risk flagged students

ACTION

Standard advising with risk score visible

Monthly check-in

Preserve capacity for high-risk

STAKEHOLDERS: Dean of Student Success (decision) | Academic Advisors (execution) | Financial Aid & Mentoring (support)

EXPECTED VALUE

EV ON TEST COHORT (n=90)

Three retention scenarios

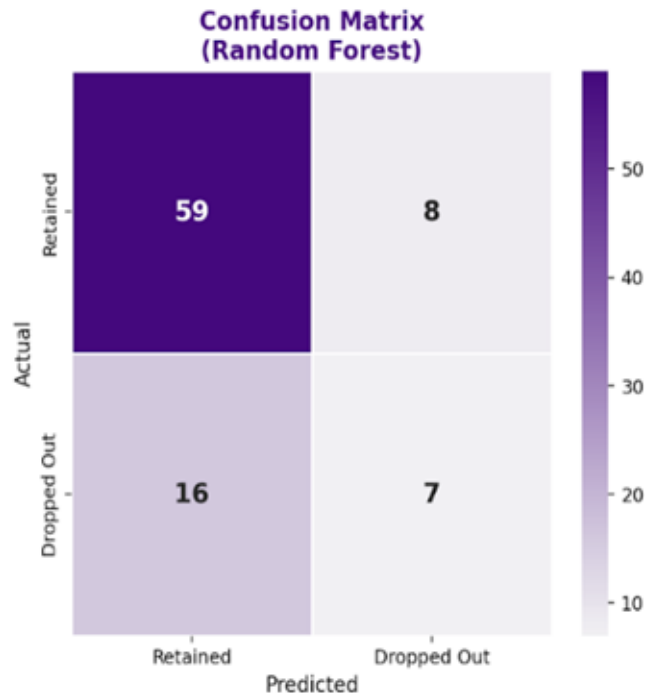
Based on 7 true positives — 15 students flagged for intervention

Conservative (5% lift) **-\$2,250**

Base (10% lift) **+\$3,000**

Optimistic (20% lift) **+\$13,500**

\$15K saved per retained student | \$500 outreach cost per flag



MEASUREMENT PLAN

TECHNICAL KPIs

Model performance

Recall, Precision, F1, AUC

Refresh every term to confirm model remains useful as a screening tool

BUSINESS KPIs

Institutional outcomes

Dropout rate in flagged cohort

Retention lift vs. baseline cohort

Cost per retained student

Tuition revenue protected

SMART TARGET: Reduce dropout in flagged cohort by 5pp within one term, with positive EV and 90%+ follow-up rate

RISKS, ETHICS & NEXT STEPS

What this analysis can and cannot do

Analytical limits. Predictive, not causal. Mental health, housing burden, and family obligations are not in the data — omitted-variable bias affects feature interpretation.

Fairness. First-gen students drop out at 35.7% vs 21.6%. Audit the model per term to ensure it does not reinforce the very disparity it's meant to address.

Governance. The model screens; advisors decide. Every flagged student passes through human review before action.

Next step: Pilot with a randomized control group inside the flagged cohort.
Scale only if retention lift and EV are demonstrated.